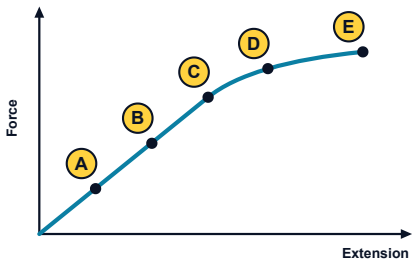


Forces and elasticity

Paper 2 · Combined and Separate
20 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 9 starred (★) questions first.

QUESTION	ANSWER	✓
1	Give three ways in which the shape of an object can be changed.	Stretching, compressing or bending ☐
2 ★	What is elastic deformation?	The object returns to its original shape when the forces are removed. ☐
3 ★	What is inelastic deformation?	The object does not return to its original shape when the forces are removed. ☐
4 ★	State how the extension of a spring depends on the force applied, below the limit of proportionality.	The extension is directly proportional to the force. ☐
5	Describe the relationship between the force on an elastic object and its extension.	As the force increases, the extension increases. ☐
6 ★	What is the limit of proportionality of an elastic object?	The point beyond which extension is no longer directly proportional to force. ☐
7	Which letter shows the limit of proportionality on this force–extension graph? 	C ☐
8	State the standard unit of force.	Newton (N) ☐
9 ★	State the standard unit of spring constant.	Newton per metre (N/m) ☐
10 ★	Define spring constant.	The force needed to stretch (or compress) a spring by 1 m. ☐

QUESTION		ANSWER	✓
11	State the standard unit of extension.	Metre (m)	☐
12 ★	State how the extension of an elastic object is calculated.	Extension = stretched length – original length	☐
13 ★	Name the energy store that increases when an elastic object is stretched or compressed.	The elastic potential energy store	☐
14 ★	Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis).	It is the gradient of the straight section.	☐
15	Describe how to find the spring constant from a graph of extension (y-axis) against force (x-axis).	It is $1 \div$ gradient of the straight section.	☐
16	State the standard unit of elastic potential energy.	Joule (J)	☐
17	State the equation linking force, spring constant and extension.	force = spring constant $\times$ extension ($F = k e$)	☐
18	Explain why more than one force is needed to stretch or compress a stationary object.	A single force would just make the object move.	☐
19	State how the work done stretching a spring compares with the elastic potential energy stored.	They are equal (if the spring is not inelastically deformed).	☐
20	State how linear and non-linear relationships look on a force–extension graph.	Linear: a straight line. Non-linear: a curve.	☐